

4. Literature Review

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Lecture 3 Recap

It's important to **summarise** research papers as you read them.

Level	Title	Description
Level 1	Annotated Bibliography	A brief summary of a paper, its relevance to your research, and your opinion of it.
Level 2	Summary	A longer and more detailed summary of the paper.
Level 3	Critique	A critical analysis of the paper.
Level 4	Peer Review	Similar to Level 3, but with a recommendation to an editor.
Level 5	Literature Survey	Covers several papers, reflection with an integrated narrative.

Lecture 3 Recap

Critical review is an important research skill to develop;

Read papers methodically and summarise them consistently.

- Always **summarise** a paper in your own words;
 - A brief synopsis (Level 1) helps in future by giving a quick overview of the content.
- Write a **structured review** on the content (Level 2/3);
 - E.g., its claimed contributions, methods, conclusions.
 - Be reflective: why did you read it, what did you learn, what would you do next?
- **Organise** your notes;
 - Reference managers, research journals, etc.

Lecture Outline

- What is a literature review?
- Searching for relevant papers.
- Writing a literature review.
- Citing literature.

What is a Literature Review?

Level	Title	Description
Level 5	Literature Survey	Covers several papers, reflection with an integrated narrative.

A literature review/survey is a **narrative summary** of many papers.

In research, “**the literature**” is a body of published works;

- Mostly used to refer to research papers;
- But may include other sources:
 - Articles from government bodies, industry whitepapers, hardware manuals, etc.
- Rather than the collective set of all scientific papers, “the literature” is usually contextual; e.g., relevant to a specific topic or research area.

What is a Literature Review?

A **lit review** (or **survey**) is a piece of writing that discusses the literature on a focused topic, with an underlying 'story'.

Typically presents a structured **summary** of the literature;

- Should **not** just be a list of paper summaries;
- Instead, should **critically reflect** on papers **individually** and **collectively**.

Reviews often present **a new way of looking** at the literature;

- E.g., by identifying trends, classifying solutions, comparing methods.
- Often, the aim is to identify a '**gap**' where research (by you) is needed.

Critical Analysis

A lit review is a **critical summary** of the literature.

Think about the literature as a whole, not just individual works:

- Identify **different approaches**;
- Identify **contradictions**;
 - E.g., contradictory findings, opposing solutions.
- Analyse the **strengths** and **weaknesses** of solutions;
- Synthesise the 'bigger picture' of the literature;
 - I.e., what is the **current state of the art**?
 - How does it all fit together?

Critical Analysis

A lit review should **reach conclusions**;

- Remember, it's not just a list of papers.
- For example:
 - Were there any surprising trends?
 - Did anything jump out as being unusual or worth further investigation?
 - Are there 'gaps' in the literature?: unsolved problems, areas for improvement, etc.

Identify the **implications** for your own work (if appropriate);

- For example:
 - Identify promising solutions;
 - Identify methods you might use;
 - Identify the key pieces of work you aim to improve on.

Why write a Literature Review?

New project: conduct a lit review to **become familiar with** a new topic;

- An important part of the research process, helping to **generate ideas** and understand how others approach your research area.
- Also important: making sure others **haven't already solved** your problem!

Writing a paper or dissertation: lit review **provides context** for the work;

- Tell the reader what others have **already done**.
- More importantly, tell the reader how your approach differs and why it is **novel**.

As a research contribution: lit reviews published as complete works;

- Useful for researchers beginning to study a new area.
- May **reflect on trends** or present a **new way of thinking** about a problem.

ACM Computing Surveys

A journal specialising in computing science literature reviews.

*“The primary purpose of the ACM Computing Surveys is to present new specialties and help practitioners and researchers stay abreast of all areas in the rapidly evolving field of computing. Computing Surveys focuses on **integrating and adding understanding** to the existing literature by publishing surveys and tutorials.”*

See: <https://csur.acm.org/>

Example from ACM Computing Surveys

L. J. Gonçalves, K. Farias, T. C. De Oliveira, and M. Scholl: **Comparison of Software Design Models: An Extended Systematic Mapping Study**. ACM Computing Surveys 52, 3 (2019).

“This article, therefore, focuses on providing a **classification** and a **thematic analysis** of studies on the comparison of software design models.”

- Surveys a **specific area**: studies comparing software design models;
- Adds **extra value**: proposes a way of classifying these studies;

Example from ACM Computing Surveys

Comparison of Software Design Models: An Extended Systematic Mapping Study

48:17

Table 5. Classification of the Primary Studies Based on Their Supported Diagrams (RQ1)

Diagram types	#Studies	Percentage	List of primary studies
Class diagram	22	39%	[S7], [S10], [S22], [S24], [S25], [S26], [S28], [S32], [S33], [S35], [S36], [S37], [S38], [S40], [S41], [S42], [S45], [S47], [S48], [S52], [S53], [S56]
Generic diagram	17	30%	[S2], [S3], [S4], [S8], [S9], [S11], [S14], [S19], [S20], [S21], [S23], [S34], [S39], [S44], [S46], [S50], [S51]
Metamodels	3	5%	[S12], [S29], [S49]
Sequence diagram	4	7%	[S30], [S31], [S35], [S55]
Statechart	3	5%	[S15], [S16], [S43]
All UML diagrams	3	5%	[S18], [S27], [S54]
Activity diagram	2	4%	[S13], [S56]
Business process model	1	2%	[S6]
Use case	1	2%	[S5]
Component and connector	1	2%	[S1]
UML profile	1	2%	[S17]
Total	58*		

*[S35] appears in two categories: Class diagram and Sequence diagram.

*[S56] appears in two categories: Activity diagram and Class diagram.

Example from ACM Computing Surveys

L. J. Gonçalves, K. Farias, T. C. De Oliveira, and M. Scholl: **Comparison of Software Design Models: An Extended Systematic Mapping Study**. ACM Computing Surveys 52, 3 (2019).

“The main results are that a majority of the primary studies (1) provide coarse-grained techniques of the comparison of general-purpose diagrams, (2) adopt graphs as principal data structure and compare software design models considering structural properties only, (3) pinpoint commonalities and differences between software design models rather than assess their similarity, and (4) propose new techniques while neglecting the production of empirical knowledge from experimental studies.”

- Also yields new insights into the research area;
 - Which might inspire **new research directions** for the field.

Systematic Review

A **rigorous** and **systematic** type of literature review;

- Clearly defined research question;
- Specifies search terms and methods for finding papers;
- Clear **inclusion** and **exclusion criteria**;
- Systematic method for **extracting relevant information**;
- Often for purposes of **meta-analysis**;
 - Synthesising findings from many studies to draw broad conclusions.
 - Mostly used in medicine for wider analysis of treatment efficacy.

Research Papers vs Review Papers

A **research paper** reports an original piece of research;

- Presenting methods, results, conclusions, etc.
- Original: e.g., a new algorithm, an experiment, a new system.

A **review paper** (or survey paper) reports on published works;

- Reviews the literature, synthesising findings from the papers.
- A **systematic review** is a special type of review paper, based on a rigorous and systematic approach.

Recap on Literature Review

A literature review summarises a body of published works;

- Has a **specific focus**:
 - Don't be too *broad*: if it's too general, it's not useful;
 - Don't be too *narrow*: it might miss important works.
- Plays many roles in the research process:
 - During investigation, when writing papers, as works themselves.

A good lit review isn't just a summary of papers:

- Presents a **critical analysis** of the literature;
- Presents **new ways of thinking about** a research area;
- Has a **narrative** that 'tells a story'.

Searching the Literature

How do you find **relevant** literature?

- Especially challenging because of the wide variety of sources!

Ask **experts** to recommend a starting point:

- Supervisors, librarians, other researchers;

Academic search engines and libraries:

- Google Scholar: <http://scholar.google.co.uk>
- ACM Digital Library: <https://dl.acm.org>
- IEEE Xplore: <https://ieeexplore.ieee.org>
- U of G Library: <https://www.gla.ac.uk/myglasgow/library>

Google Scholar

virtual reality haptics

Articles

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[CITATION] Force and touch feedback for **virtual reality**

GC Burdea, FP Brooks - 1996 - Wiley New York

☆ Cited by 1558 Related articles All 4 versions

Haptic feedback: A brief history from telepresence to virtual reality

RJ Stone - International Workshop on **Haptic Human-Computer ...**, 2000 - Springer

This paper presents a short review of the history surrounding the development of **haptic** feedback systems, from early manipulators and telerobots, used in the nuclear and subsea industries, to today's impressive desktop devices, used to support real-time interaction with ...

☆ Cited by 240 Related articles All 13 versions

Haptic glove with MR brakes for virtual reality

J Blake, HB Gurocak - IEEE/ASME Transactions On ..., 2009 - ieeexplore.ieee.org

Haptic gloves open up the world of force feedback by allowing the user to pick up and feel **virtual** objects in a natural way. In most of the existing gloves, a remote box houses a large number of actuators and sensors. Power to the glove is transmitted via cables. If the **haptic** ...

☆ Cited by 229 Related articles All 3 versions

Control law design for haptic interfaces to virtual reality

RJ Adams, B Hannaford - IEEE Transactions on control systems ..., 2002 - ieeexplore.ieee.org

The goal of control law design for **haptic** displays is to provide a safe and stable user interface while maximizing the operator's sense of kinesthetic immersion in a **virtual** environment. This paper outlines a control design approach which stabilizes a **haptic** ...

☆ Cited by 220 Related articles All 7 versions

[HTML] The value of **haptic** feedback in conventional and robot-assisted minimal invasive surgery and **virtual reality** training: a current review

OAJ Van der Meijden, MP Schijven - Surgical endoscopy, 2009 - Springer

Background **Virtual reality** (VR) as surgical training tool has become a state-of-the-art technique in training and teaching skills for minimally invasive surgery (MIS). Although intuitively appealing, the true benefits of **haptic** (VR training) platforms are unknown. Many ...

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virtual reality haptics

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1 [Towards hand-eye coordination training in virtual knee arthroscopy](#)

Shahzad Rasool, Alexei Sourin, Pingjun Xia, Bin Weng, Fareed Kagda

October 2013 VRST '13: Proceedings of the 19th ACM Symposium on Virtual Reality Software and Technology

Publisher: ACM

Bibliometrics: Citation Count: 1
Downloads (6 Weeks): 3, Downloads (12 Months): 19, Downloads (Overall): 282

Full text available: PDF

Minimally invasive arthroscopic surgery has replaced the common orthopaedic surgery procedures on joints. However it demands from surgeons to acquire very different motor-skills for using special miniature pencil-like instruments and cameras inserted through little incisions on the body while observing the surgical field on a video monitor. Training in virtual ...

Keywords: arthroscopy, virtual reality, haptics

[result highlights]

2 [The perceived roughness of resistive virtual textures: II. effects of varying viscosity with a force-feedback device](#)

Susan J. Lederman, Roberta L. Klatzky, Christine Tong, Cheryl Hamilton

January 2006 ACM Transactions on Applied Perception (TAP): Volume 3 Issue 1, January 2006

Publisher: ACM

Bibliometrics: Citation Count: 3

- 53,000 results from Google Scholar; 37,000 results from ACM DL
- What should you actually read?

Searching the Literature

How do you find good leads amongst the search results?

Citation count loosely correlates with a paper's impact;

- But obviously biases older works.

It helps to know the **top publication venues** in an area.

- Conferences and journals vary significantly in quality.
 - Generally speaking, ACM and IEEE conferences are reputable for computing science.
- The “best” conferences and journals are a good starting point.
- Skim the list of titles and look for possibly relevant papers.

Also helps to know the **top researchers and groups** in an area.

- Look for their names in the results;
- Also look at what they publish and where they publish their work.

Searching the Literature

Look at **other papers** to see what they cite:

- Follow up on interesting references;
- Look for authors that get cited frequently;
- Look for journals or conferences that get cited frequently;

Use **advanced search** tools (see next slide):

- Search for recent articles;
- Search for related articles;
- Search for papers that cite a search result;
- Search for other papers by the same authors;

Searching the Literature

Google Scholar

making sense of sensing systems

Search



Articles

About 2,980,000 results (0.13 sec)

Any time

Since 2019

Since 2018

Since 2015

Custom range...

Sort by relevance

Making sense of sensing systems: five questions for designers and researchers

[V Bellotti](#), [M Back](#), [WK Edwards](#), [RE Grinter](#)... - ... in computing **systems**, 2002 - dl.acm.org

This paper borrows ideas from social science to inform the design of novel" sensing" user-interfaces for computing technology. Specifically, we present five design challenges inspired by analysis of human-human communication that are mundanely addressed by traditional ...



Cited by 431

Related articles

All 26 versions

Searching the Literature

How do you know what phrases to search for?

- Identify **keywords** and **key concepts** in the papers you read.
 - Often in title, abstract, or keywords on front page.
- Expand your search to include **synonyms** for the core terms;
 - E.g., “virtual reality”, “VR”, “mixed reality”, “XR”, “augmented reality”.

Keep notes in your research journal about your **search terms** or **advanced search paths**;

- E.g., if you’ve looked through the list of papers that cite a certain article, you don’t want to repeat this search.

Searching the Literature

ACM & IEEE focus on computing science.

Use **advanced search** to narrow in on your research area.

- Conferences, journals
- “Sponsors”
 - SIGCHI, SIGIR, SIGAI, SIGGRAPH, etc.

Searched for acmdlSponsor:(+SIGCHI) [new search] [edit/save query] [advanced search]

Searched The ACM Full-Text Collection: 567,250 records [Expand your search to The ACM Guide to Computing Literature: 2,867,312 records] ?

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- 1** [GroupLens: an open architecture for collaborative filtering of netnews](#)
[Paul Resnick](#), [Neophytos Iacovou](#), [Mitesh Suchak](#), [Peter Bergstrom](#), [John Riedl](#)
October 1994 CSCW '94: Proceedings of the 1994 ACM conference on Computer supported cooperative work
Publisher: ACM
Bibliometrics: Citation Count: 1,400
Downloads (6 Weeks): 44, Downloads (12 Months): 619, Downloads (Overall): 10,940
Full text available: [PDF](#)
Collaborative filters help people make choices based on the opinions of other people. GroupLens is a system for collaborative filtering of netnews, to help people find articles they will like in the huge stream of available articles. News reader clients display predicted scores and make it easy for users to ...
Keywords: electronic bulletin boards, information filtering, Usenet, selective dissemination of information, collaborative filtering, netnews, social filtering, user model
[\[result highlights\]](#)
- 2** [Tangible bits: towards seamless interfaces between people, bits and atoms](#)
[Hiroshi Ishii](#), [Brygg Ullmer](#)
March 1997 CHI '97: Proceedings of the ACM SIGCHI Conference on Human factors in computing systems
Publisher: ACM
Bibliometrics: Citation Count: 1,322
Downloads (6 Weeks): 45, Downloads (12 Months): 1,267, Downloads (Overall): 16,378
Full text available: [PDF](#)
Keywords: center and periphery, tangible user interface, ambient media, augmented reality, foreground and background, graspable user interface, ubiquitous computing
[\[result highlights\]](#)
- 3** [Social information filtering: algorithms for automating “word of mouth”](#)
[Upendra Shardanand](#), [Pattie Maes](#)
May 1995 CHI '95: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems

Search Process

1. Find a good **starting point**;
 - Ask someone, use search results, etc.
 - Ideally an **important** or **recent** paper in your area.
2. Look at the **list of references** at the end;
 - If a paper title looks relevant, find it and add to your list of papers.
 - Skim the abstract and intro to help screen papers.
3. Look for **related articles** and **articles that cite** your start point;
 - The best known papers are typically a few years or decades old, so look for recent papers too, to see what your peers are **currently citing**.
4. **Repeat** the process, **branch out**, take leads from **quality** papers.

Search Process

Eventually you'll have a long list of papers...

- Screen each paper for **relevance**: title, abstract, intro, skim the rest;

Read and summarise papers as discussed in Lecture 3;

- Remember to **stay organised**;
 - Keywords, folders, etc., within your reference manager;
 - Take notes about how you think different papers fit together;
- Make sure you have **correct meta-data** for each paper;
 - Authors, title, journal name, conference name, **DOI**, etc.
 - Published papers must accurately and completely reference cited works;
 - Do it properly as you go to save time later!

Recap on Literature Search

Literature search is the time-consuming process of finding relevant literature;

- Which, eventually, you'll **summarise**.

Start with a search term, or particular paper, and **branch out**;

- Look for referenced papers to add to your list;
- Look for new keywords to search for;
- Use advanced search to find related papers, citing papers, etc.
- Take notes about what you're doing, so you don't repeat later.

Planning a Literature Review

Be clear about **why** you are conducting and writing a lit review:

- Not “*because I have to*”;
- Identify your **aims**:
 - To **better understand** an unfamiliar topic?
 - To **critically evaluate** a research area?
 - To **see what approaches have been used** to address a research problem?
 - To identify a **research gap** that you intend to explore?
 - To provide **motivation** and **background** in a research paper or dissertation?

Consider the **scope** of your review:

- What is **relevant**?
- Also, what is **not relevant**?
- Find balance between including enough work but not being too general.

Organising a Literature Review

Consider how to **structure** the summary;

- Start with **high-level organisation**:
 - What are the key high-level topics (i.e., your main headings)?
 - What order should these high-level topics be presented in?
- From there, **work down**:
 - How will you organise papers within each topic?
 - Chronological;
 - E.g., show advances, development of new approaches, changing trends, etc.
 - By theme/method/approach;
 - E.g., show and contrast related approaches, emphasise similarities or differences, etc.
- Structure is **dynamic**: be prepared to revise, try new organisation, etc.

Organising a Literature Review

Structure sometimes follows from the aims of the review;

- E.g., if reviewing advances in computer vision object identification:
 - Begin with brief chronological overview of key approaches;
 - Structure main discussion by approach type, papers in chronological order.
- E.g., if reviewing the applications of sentiment analysis algorithms:
 - Do thematic analysis to identify broad application areas;
 - Structure main discussion by theme, papers could be in chronological order or could be clustered based on similar approaches/methods/etc.

Organising a Literature Review

Structuring disparate topics;

- Sometimes necessary to discuss distinct research topics;
 - E.g., if attempting to unify two areas or draw on knowledge from different areas.
- Think of a **logical order** for each topic;
 - E.g., your intended contribution might focus on one area in particular.
- Think about how to **unify** them;
 - E.g., how will you bring these two topics together?

Writing a Literature Review

How much do you write about each paper?: “*it varies*”

Write as **little as necessary** to say as much **as you need to**;

- Be **selective** about what should and should not be discussed;
 - **Clear aims** help here: e.g., how much is needed to **support your arguments**?
- You might need to write more about certain papers;
 - E.g., if they make a particularly **important** or **relevant** contribution;
 - E.g., if you are describing a method or approach you intend to use yourself;
- Likewise, you might not need to write much for other papers;
 - E.g., if **related**, or a **minor extension**, to something already discussed;
- This comes with practice and experience;
 - You’ll get a feel for this as you read and write more papers.

Writing a Literature Review

Your **Level 1** and **Level 2** summaries are a helpful starting point;

- You have written a brief and detailed summary of each paper, use them!

Remember: a lit review is not just a list of paper summaries;

- **Critically reflect** on the literature;
- **Summarise and synthesise**: what does it all mean?;
- Construct a **narrative** that summarises the body of research;
- Use the literature to **support your arguments**;
- Think about the **aims** of the lit review – have you answered them?

Writing a Literature Review

Writing a literature review is hard work;

- For most PhD students, it's the most daunting part of thesis writing!
- Even after writing hundreds of paper summaries, constructing a good narrative is the hard part.

Writing, in general, is an **iterative** process;

- Get the first draft finished, then think about how to improve it.
- Writing (and revision) never ends, you just stop.
 - Frank Herbert: *"There is no real ending. It's just the place where you stop the story."*

Recap on Writing a Literature Review

Be clear about **why** you are writing a literature review;

- Identify **aims**, define **scope**: what will and won't you include?
- What is the **narrative** of your review: what story are you trying to tell?

Think about how to structure the review;

- Start at a high level, work down.
- Structure often follows your aims.

Not just a sequence of paper summaries;

- **Critically reflect** on them;

Citing Work

In academic writing, it is necessary to **cite** published works;

- To **identify** a source you are referring to;
 - E.g., when discussing papers in a literature review;
- To **attribute** an idea or quote to its original source;
 - E.g., when describing an idea that is not your own;

Citations use **standardised formats** to ensure each source can be properly identified and found by your readers.

- Remember: get all meta-data as you find papers, save time later!
- Ref managers (e.g., Mendeley) and tools (e.g., BibTeX) simplify this.

Citing Work

Citations are embedded in the writing, with a variety of formats;

- **Numeric** style:
 - A number in square brackets identifies the item in a numbered reference list;
 - This is the **most common format** used in computing science publications.
 - Example: “*Freeman et al. [7] argued...*”
- **Author** style:
 - Combination of first-author name and year identifies an item in the reference list;
 - Example: “*Freeman et al. (2009, 2010) argued...*”
 - Example: “*Others [Freeman, 2009] have argued...*”

Different publication venues have different citation formats;

- Use a reference manager to do it automatically;
- E.g., the ACM LaTeX template takes a BibTeX library, does everything for you.

Citing Work

Citations are **not nouns**;

- Do **not** write “*In [3], a new method was...*”
- Always name the source, even indirectly:
 - Example: “**Freeman et al.** [3] proposed a new method for...”
 - Example: “**Others** [3] proposed a new method for...”
 - Example: “**A new method** [3] was proposed for...”
- Conventions for naming authors:
 - If three or more authors, use “**et al.**”;
 - Latin for “**and others**”
 - Example: “Freeman **et al.** [3]...”
 - Otherwise, use surname of the author(s);
 - Example: “**Freeman** [3]” or “**Freeman and Brewster** [4]”

Citing Work

Add the citation at an **appropriate** place in the sentence;

- Before you add your own opinions, ideas, arguments, etc:
 - **Bad**: “Freeman et al. suggested that..., but this has not been validated **[1]**.”
 - **Good**: “Freeman et al. suggested that... **[1]**, but this has not been validated.”
 - Because “but this has not been validated” is your critique, not Freeman’s.

Citations can (and should) be used in a list;

- “Freeman et al. **[4, 8-10]**” is neater than “Freeman et al. **[4]**, **[8]**, **[9]**, **[10]**”

Caveat: always follow convention and formatting rules.

Bibliography

References appear in the **bibliography** at the end of a paper;

- A complete list with full meta-data for each cited piece of work.

Order of references is important;

- Different publication venues have their **own formatting conventions**;
- Example: Alphabetical then chronological order;
 - Jones 2016, Lee 2017, Lee 2019, Thompson 2010
- Example: Ordered by appearance in text;
 - First citation is [1], next citation is [2], etc

Again, use a **reference manager** to do this for you.

REFERENCES

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Example:

- Recent **ACM** paper;
- Numbered refs in **alphabetical then chronological** order.
- Reference format:
 - Full names, year, paper title, publication, publisher, page numbers, **DOI**.
 - **DOI**: a unique document identifier.

Lecture Summary

A literature review is a **critical analysis** of “the literature”;

- Engaging with the literature is important during the research process;
- Remember, not just a sequence of paper summaries;

Searching for **relevant** literature is a time-consuming process;

- Choose a starting point and branch out;
- Take notes as you go, stay organised, use a reference manager;

Lecture Summary

When writing a literature review:

- Have clear **aims** and define the **scope** of the review;
- **Structure** is important;
- Think about how structure and content help **achieve your aims**;
 - E.g., by adding support for an argument, by showing development of an approach.

Citing sources is an important part of academic writing;

- See “Plagiarism” in Lecture 3;
- Combine **good practice** and **convention**;
- Use reference managers and tools (e.g., BibTeX) to save time;